

CANDIDATE
NAME

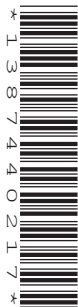
--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--



CAMBRIDGE INTERNATIONAL MATHEMATICS

Paper 2 (Extended)

0607/22

May/June 2017

45 minutes

Candidates answer on the Question Paper.

Additional Materials: Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

Do not use staples, paper clips, glue or correction fluid.

You may use an HB pencil for any diagrams or graphs.

DO NOT WRITE IN ANY BARCODES.

Answer **all** the questions.

CALCULATORS MUST NOT BE USED IN THIS PAPER.

All answers should be given in their simplest form.

You must show all the relevant working to gain full marks and you will be given marks for correct methods even if your answer is incorrect.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 40.

This document consists of **8** printed pages.

Formula List

For the equation $ax^2 + bx + c = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Curved surface area, A , of cylinder of radius r , height h . $A = 2\pi rh$

Curved surface area, A , of cone of radius r , sloping edge l . $A = \pi rl$

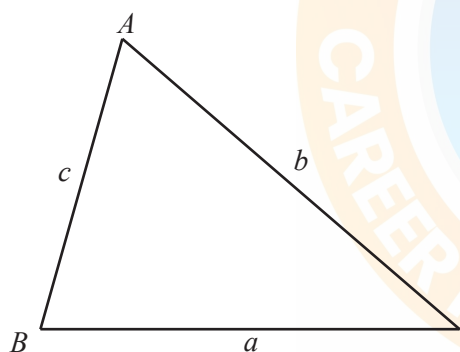
Curved surface area, A , of sphere of radius r . $A = 4\pi r^2$

Volume, V , of pyramid, base area A , height h . $V = \frac{1}{3}Ah$

Volume, V , of cylinder of radius r , height h . $V = \pi r^2 h$

Volume, V , of cone of radius r , height h . $V = \frac{1}{3}\pi r^2 h$

Volume, V , of sphere of radius r . $V = \frac{4}{3}\pi r^3$



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}bc \sin A$$

Answer **all** the questions.

- 1 (a) Write 5.30987 correct to 3 decimal places.

..... [1]

- (b) Write 0.003 648 9 correct to 3 significant figures.

..... [1]

- 2 These are the number of points *The Storm* have scored in their last 20 basketball matches.

28	33	49	37	26
54	46	48	53	34
26	17	46	41	52
48	37	30	45	53

- (a) Construct an ordered stem and leaf diagram to show these scores and complete the key.

.....	
.....	
.....	
.....	
.....	

Key | = 53 [3]

- (b) Find the median score.

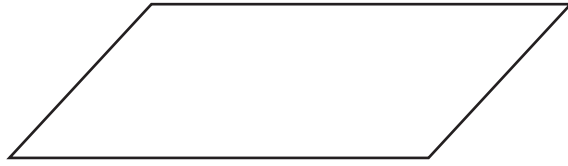
..... [1]

- 3 Factorise completely.

$$6x^2 - 2x$$

..... [2]

4

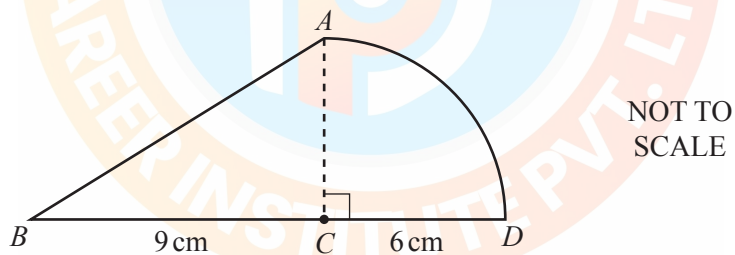


Complete this statement for the parallelogram shown.

This shape has lines of symmetry and rotational symmetry of order [2]

5 Simplify $4(2x - 1) - 3(x - 2)$.

6



AD is an arc of a circle, centre C , and BCD is a straight line.
 $BC = 9\text{ cm}$, $CD = 6\text{ cm}$ and angle $ACD = 90^\circ$.

Find the total area of the shape $ABCD$.
 Give your answer in terms of π .

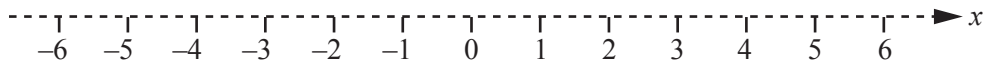
..... cm^2 [3]

7 $3x + 2 \geq 5x - 6$

(a) Solve the inequality.

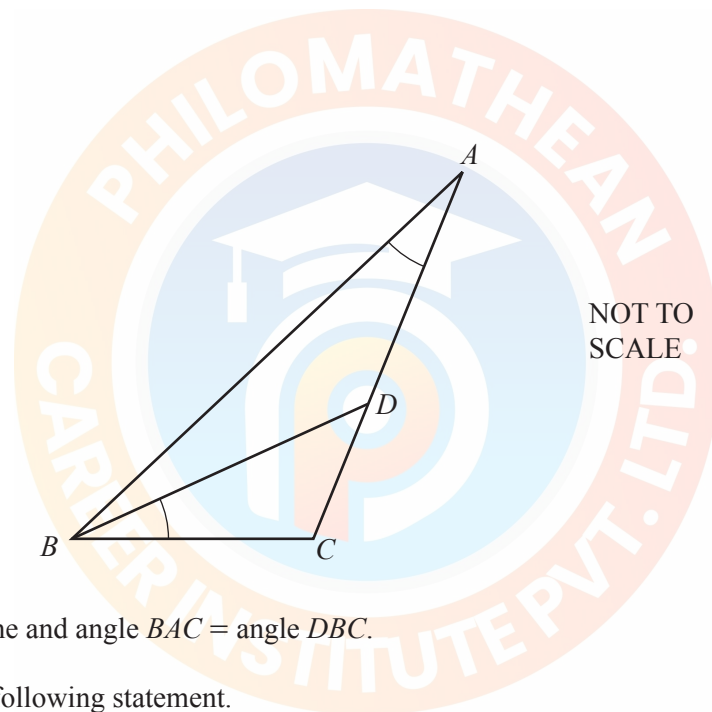
..... [2]

(b) Show your solution to **part (a)** on this number line.



[1]

8



ADC is a straight line and angle $BAC = \text{angle } DBC$.

(a) Complete the following statement.

Triangle ACB is similar to triangle

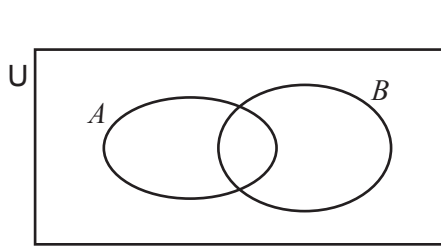
[1]

(b) $BC = 6 \text{ cm}$ and $CD = 4 \text{ cm}$.

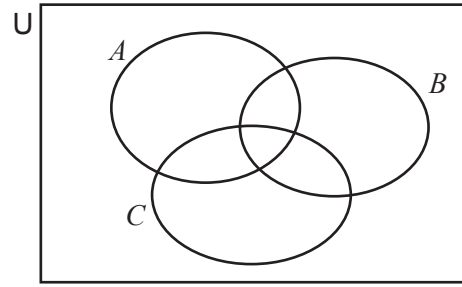
Calculate the length AC .

$AC = \dots\dots\dots \text{ cm}$ [2]

- 9 (a) In each diagram, shade the region indicated.



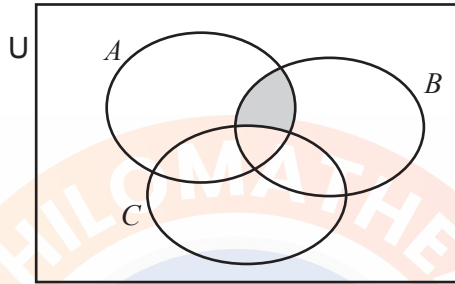
$$A \cap B'$$



$$(A \cup C) \cap B'$$

[2]

- (b) Use set notation to describe the shaded region.



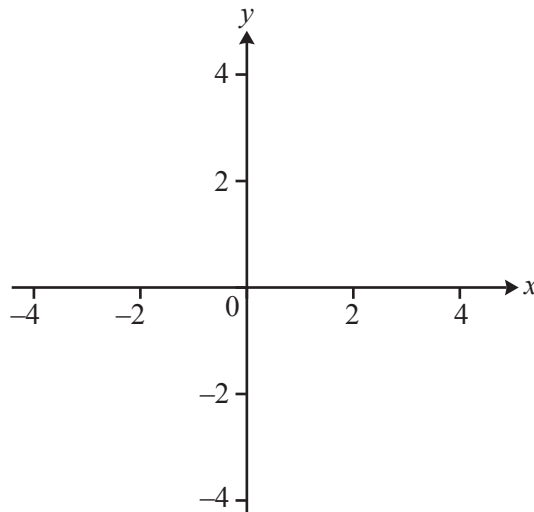
[1]

- 10 Expand the brackets and simplify.

$$(2x - 3y)(3x - 4y)$$

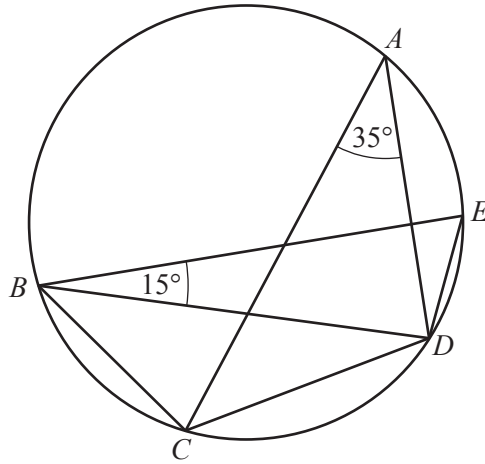
[3]

- 11 Sketch the graph of $y = |x + 2|$.



[3]

12

NOT TO
SCALE

A, B, C, D and E are points on the circle.
Angle $CAD = 35^\circ$ and angle $EBD = 15^\circ$.

Find

(a) angle CBD ,(b) angle CDE .Angle $CBD = \dots\dots\dots [1]$ Angle $CDE = \dots\dots\dots [1]$

13

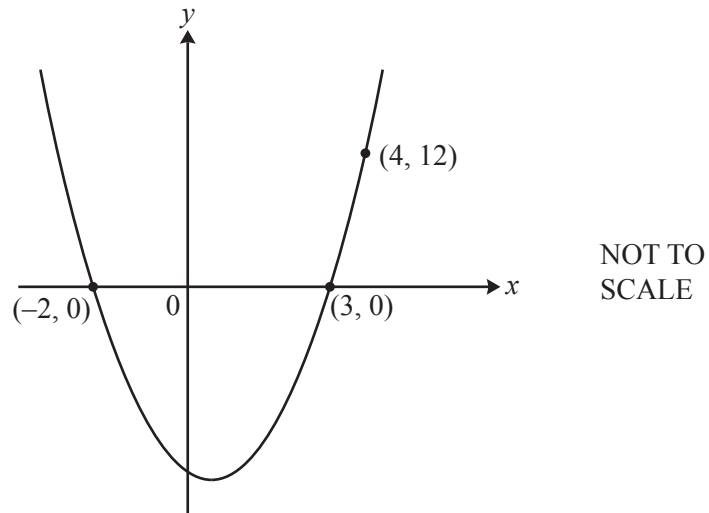
$p = 5 + 2\sqrt{3}$

$q = 5 - 2\sqrt{3}$

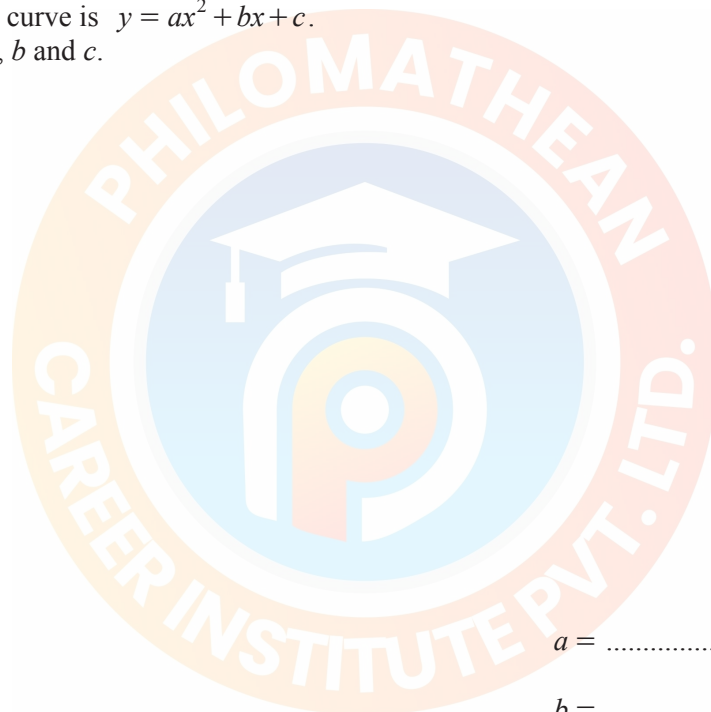
Find $p^2 - q^2$, writing your answer in its simplest form. $\dots\dots\dots [3]$ 14 Find the value of x when $5 \log 2 - \log 8 = \log x$. $x = \dots\dots\dots [2]$

Question 15 is printed on the next page.

15



The equation of this curve is $y = ax^2 + bx + c$.
Find the values of a , b and c .



$a =$

$b =$

$c =$ [3]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge International Examinations Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cie.org.uk after the live examination series.

Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.